



SUJET STATAPP 322 - 2026-2027

Sujet proposé par : Präg Patrick - patrick.prag@ensae.fr
Organisme : CREST
Catégorie : Sociologie Quantitative
Sujet :
Social status and health. Analysis of longitudinal survey data

Description du sujet :

It is widely known that people who are socioeconomically better off are also healthier and live longer. It is more puzzling why people who think they are better off socioeconomically are healthier than those who think they are worse off, even when comparing individuals with the same income and occupation. Social scientists have interpreted this pattern in different ways: some argue it shows that people's own perceptions of their social position directly shape their health, while others argue that traditional objective measures like income or occupation no longer fully capture social inequality in modern societies.

This StatApp takes a fresh look at the association between perceived social status and health using a rich longitudinal data set of older English people. The English Longitudinal Study of Aging (ELSA) has interviewed the same ca. 10,000 households (individuals aged 50+) every other year since 2002, asking about wealth and income as well as perceived socioeconomic status and health.

Modeling this association with random- and fixed-effects models will generate new insights, since ELSA's objective socioeconomic status measures are exceptionally rich, including wealth and a partner's resources over time, and its longitudinal design lets us control for unobserved time-constant confounders.

A previous StatApp using this approach found that social status and health are robustly correlated: people who think they are better off socially are healthier, even controlling for time-constant confounders (published as Coustaury et al. 2023). This StatApp goes a step further, asking whether an increase in status perceptions benefits health as much as a decrease harms it, using a recently developed method that estimates the effects of increases and decreases separately rather than assuming they are equal and opposite (Allison 2019).

This StatApp is based on a collaborative research project by Patrick Präg (CREST/ENSAE), Asri Maharani (University of Manchester), and Lindsay Richards (University of Oxford).

Statistical software will be Stata or R, depending on the students' preferences.

At the end of this StatApp, you will have:

- An excellent understanding of longitudinal data analysis using random- and fixed-effects estimators, including recent decomposition approaches for the fixed-effects estimator
- Hands-on experience analyzing real-world longitudinal survey data, including data preparation and handling missing information
- An excellent understanding of the English Longitudinal Study of Aging, one of a family of aging studies worldwide (similar data sets include the Survey of Health, Aging and Retirement in Europe (SHARE), of which France is a part, and the US Health and Retirement Study (HRS)), giving you the tools to answer a wide range of research questions about the population aged 50+

Données du sujet :

Steptoe et al. (2013) is a brief, yet older overview of the data to be used. Additional information about the data can be found here: <https://elsa-project.ac.uk>. The data is available here: <https://doi.org/10.5255/UKDA-SN-5050-30>.

Modèles envisagés du sujet :

Random- and fixed-effects models, including a recently developed asymmetric fixed-effects decomposition approach (Allison 2019) to separately estimate the effects of increases and decreases in perceived status. Statistical software: Stata or R, depending on student preference.

Bibliographie du sujet :

Allison, Paul D. 2019. "Asymmetric Fixed-Effects Models for Panel Data." *Socius* 5:1-12. doi: 10.1177/2378023119826441. In this article, Allison introduces the asymmetric fixed-effects estimator we are planning to use in this StatApp.

Coustaury, Camille, Elias Jeannot, Adele Moreau, Clotilde Nietge, Asri Maharani, Lindsay Richards, and Patrick Präg. 2023. "Subjective Socioeconomic Status and Self-Rated Health in the English Longitudinal Study of Aging. A Fixed-Effects Analysis." *Social Science & Medicine* 336(116235):1-9. doi: 10.1016/j.socscimed.2023.116235. This is a journal article based on a previous StatApp that will give you a good introduction into the substantive topic of this StatApp.

Steptoe, Andrew, Elizabeth Breeze, James Banks, and James Nazroo. 2013. "Cohort Profile: The English Longitudinal Study of Ageing." *International Journal of Epidemiology* 42(6):1640-1648. doi: 10.1093/ije/dys168. This article is the standard introduction to the data we are going to use. Despite the article being old, it gives you a good overview of the data.

Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. MIT Press.

Wooldridge, Jeffrey M. 2016. *Introductory Econometrics. A Modern Approach*, 6th ed. Cengage. These are standard econometric textbooks that I like, but any other econometric text will probably work, too.

Notation du sujet :**Commentaires :**